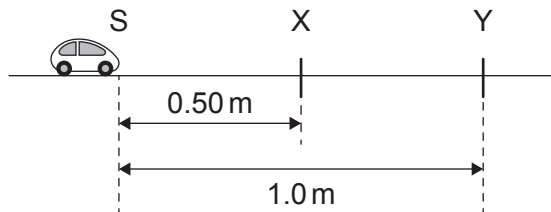


PAPER 1 / QUESTION 21

5 November 2025 / TZ1 (file label) / HL Paper 1A / original question 2 / 8825-6707

- 21.** A cart starts from rest at point S and moves in a straight line with a constant acceleration. Two points X and Y are 0.50 m and 1.0 m from S.



The cart passes X with speed v_X and passes Y with speed v_Y . What is $\frac{v_Y}{v_X}$?

- A. $\sqrt{2}$
- B. 2
- C. $2\sqrt{2}$
- D. 4